

Volume 6 – AI Decision Engine Design Document

AI Quant Trading Platform

Version: 1.0

Document Type: AI Decision Engine Design

Classification: Core System

1. Purpose

The AI Decision Engine is the central intelligence layer of the platform.

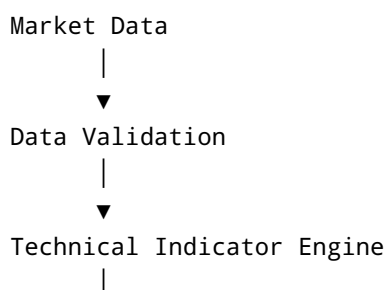
Its responsibility is to continuously evaluate every supported cryptocurrency, calculate the probability of favorable trading opportunities, estimate associated risks, and recommend actions based on configurable strategies and risk policies.

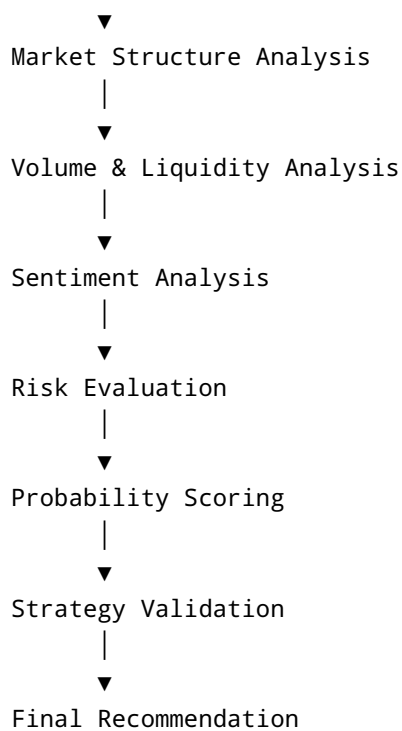
The engine provides decision support and automation. It does not predict the future or guarantee profitable trades.

2. Objectives

- Analyze all supported trading pairs continuously
 - Combine technical, market, and contextual signals
 - Score every opportunity
 - Filter low-quality setups
 - Produce explainable recommendations
 - Adapt to changing market conditions
 - Support automated and manual trading modes
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3. AI Processing Pipeline





4. Input Sources

Market Data

- Price
- OHLCV
- Order Book
- Recent Trades
- Volume
- Spread
- Funding Rate
- Open Interest
- Liquidation Data

Technical Indicators

- EMA
- SMA
- RSI
- MACD
- ATR
- Bollinger Bands
- VWAP
- ADX
- SuperTrend
- Stochastic RSI
- CCI

- ROC
- MFI
- OBV

Market Structure

- Higher High / Higher Low
- Lower High / Lower Low
- Break of Structure (BoS)
- Change of Character (ChoCH)
- Support / Resistance
- Supply / Demand Zones
- Order Blocks
- Liquidity Zones

Contextual Signals

- BTC market direction
- ETH market direction
- Stablecoin dominance
- Volatility regime
- Session timing
- Economic news
- Exchange maintenance status

5. AI Evaluation Stages

Stage 1 – Data Integrity

Reject incomplete or stale market data.

Stage 2 – Liquidity Filter

Reject illiquid markets.

Stage 3 – Volatility Filter

Reject markets outside configured volatility limits.

Stage 4 – Trend Detection

Determine:

- Bullish
- Bearish
- Sideways

Stage 5 – Momentum Evaluation

Measure trend strength and continuation probability.

Stage 6 – Multi-Timeframe Confirmation

Evaluate:

- 5m
- 15m
- 1h
- 4h
- 1d

Signals with conflicting higher-timeframe direction may be downgraded or rejected.

Stage 7 – Strategy Validation

Apply configured strategy rules.

Stage 8 – Risk Validation

Ensure the proposed trade satisfies all configured risk limits.

Stage 9 – Recommendation Generation

Output:

- Buy
 - Sell
 - Hold
 - Ignore
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6. Confidence Scoring

Each candidate receives a normalized score (0–100).

Example factors:

- Trend alignment
- Momentum
- Volume confirmation
- Market structure
- Liquidity
- Volatility
- Risk profile compatibility
- Higher timeframe agreement
- Context filters

Example thresholds:

- 90–100: Eligible for automation (if enabled)
- 75–89: High-quality watchlist
- 50–74: Monitor
- Below 50: Ignore

Thresholds are configurable by strategy and user risk profile.

7. Explainable Decisions

Every recommendation must include human-readable reasoning.

Example:

- Trend aligned across 1h and 4h
- Volume above rolling average
- RSI confirms momentum
- Breakout above resistance
- Risk-reward exceeds configured minimum

Users should understand why a recommendation was generated.

8. Market Regime Detection

The engine classifies current conditions:

- Strong Bull Trend
- Weak Bull Trend
- Strong Bear Trend
- Weak Bear Trend
- Sideways Range
- High Volatility
- Low Volatility
- News-Driven Market

Strategies may enable or disable themselves depending on the detected regime.

9. Multi-Timeframe Logic

Lower timeframe signals are validated against higher timeframe context.

Example policy:

- Entry on 5m

- Confirmation on 15m
- Trend validation on 1h
- Direction confirmation on 4h

Daily timeframe acts as macro context.

10. Strategy Compatibility Layer

Each recommendation is evaluated against:

- Trend Following
- Breakout
- Pullback
- Momentum
- Swing
- Scalping
- Grid
- DCA
- Custom Strategies

The engine may reject otherwise valid signals if they conflict with the active strategy.

11. Risk Awareness

Before recommending execution, the engine considers:

- Portfolio exposure
- Maximum open positions
- Daily loss limit
- Position sizing rules
- Drawdown state
- Correlation with existing positions

The AI cannot bypass configured risk policies.

12. Recommendation Output

Each recommendation includes:

- Symbol
- Direction
- Confidence
- Risk Level
- Entry Zone
- Stop Loss Suggestion

- Take Profit Targets
 - Strategy Name
 - Supporting Factors
 - Timestamp
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13. Learning & Feedback

The platform stores outcomes for continuous evaluation.

Captured data:

- Recommendation
- Executed or ignored
- Final trade outcome
- Market regime
- Strategy used
- Confidence score
- Risk score

Historical performance supports future model improvements but does not automatically modify production behavior without validation.

14. Safety Principles

The AI must never:

- Ignore configured risk limits
 - Submit duplicate orders
 - Recommend trades with invalid data
 - Execute when exchange connectivity is unavailable
 - Override emergency stop
 - Hide uncertainty
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15. Performance Targets

- Continuous market evaluation
 - Low-latency scoring
 - Horizontal scalability
 - Deterministic decision pipeline
 - High observability through metrics and logs
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16. Future Enhancements

Potential future capabilities:

- Adaptive weighting based on validated performance
 - Ensemble models
 - Reinforcement learning research environment
 - Cross-exchange opportunity analysis
 - On-chain analytics integration
 - Advanced news and sentiment models
 - Personalized strategy optimization (subject to user approval)
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17. Design Principles

- Decisions must be explainable.
- Risk management has higher priority than opportunity detection.
- Strategies define behavior; AI enhances evaluation.
- Recommendations are based on probabilities, not certainty.
- Every automated action must be auditable.
- The engine should fail safely when data quality or system health is insufficient.